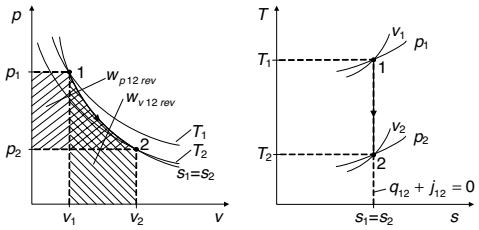
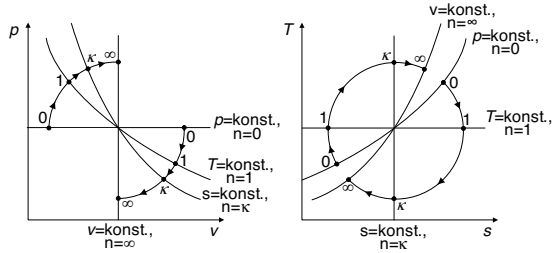
Ideal gas	$s =$ **constant: Isentropic state change**	$n =$ **constant: Polytropic state change**
Therm. state change $p\,V = n\,R\,T$ $R = 8.314\,\mathrm{J/mol\,K}$ $p\,V = m\,R_i\,T$ $R_i = \dfrac{R}{M_i}$ $p\,v = R_i\,T, \quad \rho = \dfrac{p}{R_i\,T}$ $R_i = c_p - c_v, \quad \kappa = \dfrac{c_p}{c_v}$ $c_v = \dfrac{R_i}{\kappa - 1}, \quad c_p = \dfrac{\kappa\,R_i}{\kappa - 1}$	 $s = \text{const}: \quad p_1\,v_1^\kappa = p_2\,v_2^\kappa = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} = \left(\dfrac{v_1}{v_2}\right)^{\kappa-1}$ $w_{v\,12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$ $w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \text{etc. (replace)}$	 Polytrope: $pv^n = \text{konst.}$ Isobare: n=0, $pv^0 = \text{konst.}$ Isotherme: n=1, pv = konst. Isentrope: n=κ, $pv^\kappa = \text{konst.}$ Isochore: n=∞, $pv^\infty = \text{konst.}$ $n = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{v_1}{v_2}} = \dfrac{\ln\frac{p_2}{p_1}}{\ln\frac{p_2}{p_1} - \ln\frac{T_2}{T_1}}$ $n = \text{const}: \quad p_1\,v_1^n = p_2\,v_2^n = \text{const.} \quad \Rightarrow \quad \dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{n-1}{n}} = \left(\dfrac{v_1}{v_2}\right)^{n-1}$ $w_{v\,12} = \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$ $w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc. (replace, } n \neq 1)$
1st LT closed system $q_{12} + w_{12} = u_2 - u_1$ Rev. volume change work: $w_{v\,12\,rev} = -\int_1^2 p\,\mathrm{d}v$ $w_{v\,12} = w_{v\,12\,rev} + j_{12}$ Overall work: $w_{12} = w_{v\,12} + w_{W\,12} + w_{el\,12}$	$q_{12} + w_{12} = u_2 - u_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{12} = w_{v12\,rev} = u_2 - u_1 = c_v\,(T_2 - T_1)$ $w_{v12\,rev} = \dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$	$q_{12} + w_{12} = u_2 - u_1 = c_v\,(T_2 - T_1)$ $w_{12} = w_{v\,12} = \dfrac{R_i}{n - 1}\,(T_2 - T_1) = c_v\,\dfrac{\kappa - 1}{n - 1}\,(T_2 - T_1) =$ $= \dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{12} = w_{v12\,rev}$ $q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$
1st LT open system $q_{12} + w_{t\,12} = h_2 - h_1$ Rev. pressure change work: $w_{p\,12\,rev} = \int_1^2 v\,\mathrm{d}p$ Technical work: $w_{t\,12} = \int_1^2 v\,\mathrm{d}p + j_{12}$	$q_{12} + w_{t\,12} = h_2 - h_1$ Isentropic means adiabatic ($q_{12} = 0$) and reversible ($j_{12} = 0$): $w_{t\,12} = w_{p\,12\,rev} = \kappa\,w_{v\,12\,rev} = h_2 - h_1 = c_p\,(T_2 - T_1)$ $w_{p\,12\,rev} = \kappa\,\dfrac{p_1\,v_1}{\kappa - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{\kappa-1} - 1\right] = \kappa\,\dfrac{R_i\,T_1}{\kappa - 1}\left[\left(\dfrac{p_2}{p_1}\right)^{\frac{\kappa-1}{\kappa}} - 1\right] = \text{etc.}$	$q_{12} + w_{t\,12} = h_2 - h_1 = c_p\,(T_2 - T_1)$ $w_{t\,12} = w_{p\,12} = n\,w_{v\,12} = n\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$ Rev. SC: $w_{p\,12\,rev} = n\,w_{v\,12\,rev} = \dfrac{n\,R_i}{n - 1}\,(T_2 - T_1) = \text{etc.}$ $q_{12\,rev} = c_v\,\dfrac{n - \kappa}{n - 1}\,(T_2 - T_1) = \dfrac{n - \kappa}{\kappa - 1}\,\dfrac{p_1\,v_1}{n - 1}\left[\left(\dfrac{v_1}{v_2}\right)^{n-1} - 1\right] = \text{etc.} \quad (n \neq 1)$
2nd LT / entropy $\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T}$ $\mathrm{d}s = \dfrac{\mathrm{d}u + p\,\mathrm{d}v}{T} = \dfrac{\mathrm{d}h - v\,\mathrm{d}p}{T}$	$\mathrm{d}s = \dfrac{\mathrm{d}q + \mathrm{d}j}{T} = 0 \Rightarrow s_2 - s_1 = 0$	$s_2 - s_1 = c_v\,\dfrac{n - \kappa}{n - 1}\,\ln\dfrac{T_2}{T_1} \quad (\text{for } n \neq 1)$ $s_2 - s_1 = c_v\,\ln\dfrac{T_2}{T_1} + R_i\,\ln\dfrac{v_2}{v_1} = c_p\,\ln\dfrac{T_2}{T_1} - R_i\,\ln\dfrac{p_2}{p_1}$